

Royal Rao

Senior Developer

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Summary

Senior Developer with a strong background in full-stack development and AI-driven automation. Experienced in building and deploying scalable web applications across frontend, backend, and workflow automation, with hands-on work in TypeScript, React, Python, n8n, and Google Cloud. Focused on delivering efficient, data-driven solutions that improve operations and support business growth.

Technical Skills

Programming Languages: Python, TypeScript, JavaScript, Go, SQL, C/C++

Web & Frameworks: React / React Native, Next.js, Node.js, FastAPI, HTML, CSS, RESTful APIs, Webhooks, Redis, Apache Kafka, Progressive Web Apps (PWA)

AI & Automation: n8n, OpenAI API, Ollama, LangChain, TensorFlow, scikit-learn, OpenCV, ML & AI automation pipelines

Databases: PostgreSQL, MongoDB, VectorDB, SAP

Cloud & Security: Google Cloud (VM, Load Balancer, DNS, SSL), Microsoft Azure, Cloudflare (WAF, bot mitigation), Meta / WhatsApp Business Platform

Data & Analytics: Power BI, Tableau, Pandas, NumPy, Matplotlib, Excel, Google Sheets

Tools: Git/GitHub, Docker, Postman, VS Code, TablePlus, Google Firebase

Experience

Senior Developer | Aug 2025 – Present

UK Telecom Distribution Pvt. Ltd. | New Delhi, India

- Built and deployed AI automation workflows in n8n, cutting manual work by over 40%.
- Designed an AI-powered WhatsApp support system using a local LLM (Ollama) to auto-resolve technical and non-technical user queries.
- Integrated the WhatsApp Business Platform via the Meta (Facebook) Developer console with multilingual message templates.
- Developed the B2B eSIM web platform (esim-api.com, iotsimwhole.com, tracker.esim-api.com) with a TypeScript backend, React frontend, Redis, and Kafka.
- Built a cross-platform eSIM mobile app using React Native and Go, integrated with provisioning APIs.
- Provisioned Google Cloud infrastructure (VM, load balancer, DNS, SSL) and configured Cloudflare rules to block bots and threats.
- Handled PostgreSQL databases end-to-end – table creation, schema migrations, and query tuning via TablePlus.
- Led all client-facing work – meetings, onboarding, and API setups – and drove end-to-end API testing and webhook integration using Postman.

Analyst | Aug 2024 – Aug 2025

Microtek New Technologies Pvt. Ltd. | Himachal Pradesh, India

- Automated databases, MIS reports, and KPI dashboards (Sales, Purchase, BOM Costing, NPD) using SAP, Google Sheets, and Power BI.
- Coordinated 10+ cross-functional projects with project managers and R&D, Quality, and Production teams, shortening NPD cycles and reducing defect rates.
- Built a real-time web app (HTML, CSS, JavaScript) to track PDCL Lab test runs with a completion-time estimation algorithm.
- Presented data insights and reports to the CMD and senior management to support faster decision-making.

Python Developer / Data Analyst | Aug 2023 – Aug 2024

Extech Digital | Mohali, Punjab

- Developed and maintained Python applications for data processing and reporting, emphasizing clean, reusable, and scalable code.
- Collected, cleaned, and prepared datasets using Python, SQL, and Excel for analysis and reporting.
- Built interactive visualizations in Power BI and Google Sheets to support business insights.
- Maintained and delivered reports with 98% on-time accuracy.

Integrated Telecom Networks and Cybersecurity Trainee | Jul 2023

ALTTC, BSNL | Uttar Pradesh, India

- Completed industrial training on integrated telecom and data networks, covering network architecture, protocols, and cybersecurity best practices.

Education

Master of Technology, Data Engineering | 2024 – 2026

IIT Jodhpur | Jodhpur, India

- Focused on designing and building systems to collect, store, and analyze large-scale data.

Bachelor of Technology, Electronics and Communication Engineering | 2020 – 2024

Guru Jambheshwar University of Science and Technology | Hisar, India

Projects

AI & Automation Projects

- **WhatsApp Support AI Automation for Client** – Built an end-to-end AI-powered WhatsApp support bot using a local LLM (Ollama/Qwen2.5 7B) with Python libraries (LangChain, FastAPI); automated customer query handling with context-aware responses and conversation memory. Developed a companion Progressive Web App (PWA) admin dashboard for real-time conversation monitoring, analytics, and bot configuration – installable on any device with offline support.
- **AI Predictive Models:** Movie Rating, Credit Card Fraud Detection, and Stock Price Prediction, built with scikit-learn and TensorFlow.

Full-Stack, Web & Mobile Projects

- **eSIM Mobile App** – Cross-platform app (React Native + Go) with secure provisioning and OTA activation, integrated with backend APIs.
- [esim-api.com](#), [iotesimwholesale.com](#), [tracker.esim-api.com](#) – B2B eSIM web platform (TypeScript, React, Redis, Kafka).
- **PDCL Lab Test Run Webpage** – Real-time tracking with a completion-time estimation algorithm.

Data & Dashboard Projects

- MIS Reports and Presentations; Madhav Ecommerce Dashboard.
- Automated KPI dashboards across Sales, Purchase, BOM Costing, NPD, Quality, and R&D.

M.Tech Projects

- **SignSense** – Real-time webcam-based sign language to text recognition using OpenCV and TensorFlow.
- **Air Quality Index (AQI) Forecasting & Analysis** – Azure Data Engineering Pipeline.
- **Enhanced Diabetic Retinopathy Diagnosis** – CNN-Transformer architecture for multi-class classification, with a GUI and clinical recommendations.

B.Tech Projects

- Mobile Signal Jamming Circuit; Wireless Power Transmission.

Certifications

- (Industrial Training) Integrated Telecom/Data Network & Cyber Security – ALTTC, BSNL
- Data Science Internship (Online) – Codsoft, Internpe
- C/C++ – Aptech Learning
- Python – LUDIFU, Excellence Technology
- Power BI – LUDIFU
- AI For India 2.0 – GUVI (Skill India Digital)
- Campus Recruitment Training cum Soft Skills Development Program (CRT) – Rubicon Skill Development Pvt. Ltd.
- Personality Development Course – Excellence Technology
- AWS – Simplilearn

Portfolio

Online Portfolio – a comprehensive showcase of web development projects, data analytics dashboards, and live test monitoring applications: github.com/dev-royalrao/PORTFOLIO